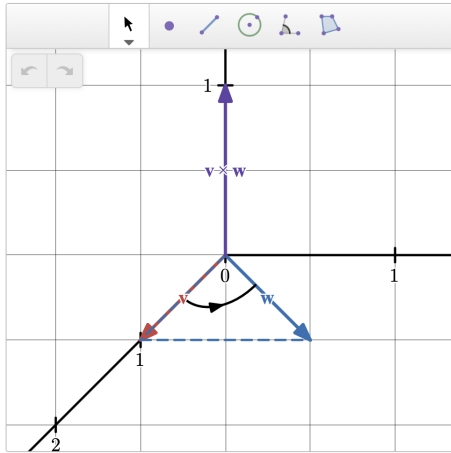


Lecture 3. A1 – Vectors and Vector Operations.

A. Cross Products. Let us talk about another useful way to multiply 3D vectors.



Let $\vec{v}$ and $\vec{w}$ be 3D vectors. Their **cross product** $\vec{v} \times \vec{w}$ is defined as follows.

- it points orthogonally to $\vec{v}$ and $\vec{w}$ in direction given by the righthand rule
- its length is $\|\vec{v} \times \vec{w}\| =$

$\vec{i}, \vec{j}, \vec{k}$ is really physics notation.
Mathematicians might prefer $\vec{e}_1, \vec{e}_2, \vec{e}_3$.

In other words, its length is the area of the parallelogram formed by $\vec{v}$ and $\vec{w}$.

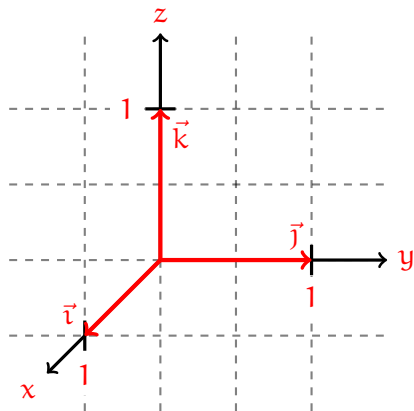
It has some unusual properties.

Let $\vec{v}$ and $\vec{w}$ be 3D vectors. Then:

self annihilating: $\vec{v} \times \vec{v} =$

anti-commutative: $\vec{v} \times \vec{w} =$

Example 1. Consider the special vectors:



$\vec{i} =$

$\vec{j} =$

$\vec{k} =$

Compute: $\vec{i} \times \vec{j} =$

$\vec{i} \times \vec{k} =$

$\vec{j} \times \vec{k} =$

$\vec{i} \times (\vec{i} \times \vec{j}) =$

$(\vec{i} \times \vec{i}) \times \vec{j} =$

Oh my god. You **cannot** freely move parentheses around? That's SO messed up. This is referred to as the **failure** of associativity: $(\vec{v} \times \vec{w}) \times \vec{r} \neq \vec{v} \times (\vec{w} \times \vec{r})$. Associativity is all about moving parentheses around.

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Example 2. Any 3D vector can be written in terms of $\vec{i}, \vec{j}, \vec{k}$ as follows.

$$\langle a, b, c \rangle =$$

Use this to compute: $\langle 1, 2, 0 \rangle \times \langle 2, 0, 0 \rangle$

Solution. First we write in terms of $\vec{i}, \vec{j}, \vec{k}$.

$$\langle 1, 2, 0 \rangle \times \langle 2, 0, 0 \rangle =$$

Next we distribute, careful to preserve order.

$$\dots =$$

Next we move scalars to the front.

$$\dots =$$

Next we apply the self-annihilating and anti-commutativity properties.

$$\dots =$$

$$\dots =$$

distributivity:

$$(\vec{v} + \vec{w}) \times \vec{r} = \vec{v} \times \vec{r} + \vec{w} \times \vec{r}$$

$$\vec{v} \times (\vec{w} + \vec{r}) = \vec{v} \times \vec{w} + \vec{v} \times \vec{r}$$

commutativity with scalars:

$$(c\vec{v}) \times \vec{w} = \vec{v} \times (c\vec{w}) = c(\vec{v} \times \vec{w})$$

B. Computing Cross Products. So far cross-products seem tough to compute.

Is there not a magic formula?

The cross-product $\langle a_1, a_2, a_3 \rangle \times \langle b_1, b_2, b_3 \rangle$ equals the **determinant**:

$$\begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

which means it equals:

$$+ \begin{vmatrix} \vec{i} & a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} - \begin{vmatrix} \vec{j} & a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} + \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} \vec{k}$$

We call a rectangular array of entries a matrix. The determinant of a 2 (rows) by 2 (columns) matrix is written with notation that looks like the absolute value of the matrix, though we are not taking an absolute value, but are computing:

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

and this is what we are computing thrice as part of calculating the cross product.

Example 3. Find a nonzero vector orthogonal to $\vec{v} = \langle 4, 5, 6 \rangle$ and $\vec{w} = \langle 7, 8, 10 \rangle$.

Solution. The cross product will produce such a vector.

We set it up as a determinant.

$$\vec{v} \times \vec{w} =$$

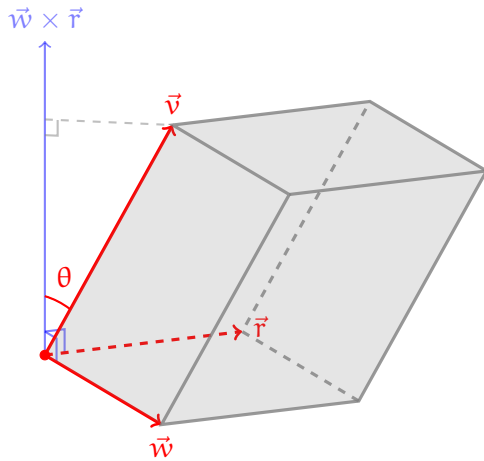
From now on, if you ever need a vector orthogonal to two other 3D vectors, cross products better leap into your mind!

C. **Scalar Triple Product.** We execute the dot and cross product in succession.

The **scalar triple product** of 3D vectors $\vec{v}$, $\vec{w}$, $\vec{r}$ is:

It has an important geometric meaning. Each pair of vectors from $\vec{v}$, $\vec{w}$, $\vec{r}$ forms a parallelogram, and together they form an object called a **parallelepiped**.

Say that three times fast. You can think of it as a slanted cube.



The base of this parallelepiped is a parallelogram formed by $\vec{w}$ and $\vec{r}$, so has area:

The height of the parallelepiped has value:

Therefore the volume of the parallelepiped is:

Or, using that $\vec{x} \cdot \vec{y} = \|\vec{x}\| \|\vec{y}\| \cos \theta$, can be written as:

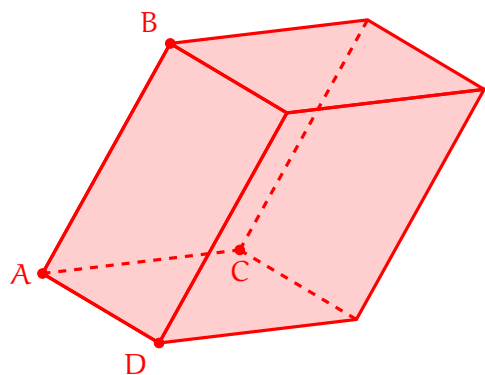
Dot products can be negative, but we insist that volume be nonnegative.

So, we make a small adjustment by introducing absolute values.

The parallelepiped formed by $\vec{v}$, $\vec{w}$, $\vec{r}$ has:

[volume] =

Example 4. Find the volume of the parallelepiped with vertex $A(1, 0, 3)$ adjacent to vertices $B(2, 2, 6)$, $C(5, 5, 9)$, $D(8, 8, 13)$.



This is probably not even close to how this parallelepiped actually looks in xyz -space. Nonetheless we make a sketch because it helps organize our thoughts. And god knows I need help with that.